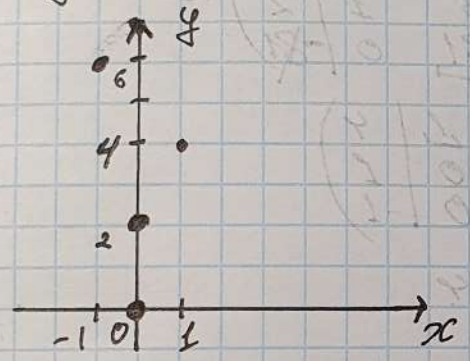
Д/З.

Макаровский ЗВ 2003_2
К 03.11.2023

№3 (1, 2)

1)

x	1	1	0	0	-1	- фактор
y	4	4	0	2	6	- отклик.



2) МНК

$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2$$

$$x = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & -1 & 1 \end{pmatrix} \qquad y = \begin{pmatrix} 4 \\ 4 \\ 0 \\ 2 \\ 6 \end{pmatrix}$$

$$x^T x = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & -1 & 1 \end{pmatrix} = \begin{pmatrix} 5 & 1 & 3 \\ 1 & 3 & 1 \\ 3 & 1 & 3 \end{pmatrix}$$

$$x^T y = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 4 \\ 4 \\ 0 \\ 2 \\ 6 \end{pmatrix} = \begin{pmatrix} 16 \\ 2 \\ 14 \end{pmatrix}$$

$$\begin{cases} 5\beta_1 + \beta_2 + 3\beta_3 = 16 \\ \beta_1 + 3\beta_2 + \beta_3 = 2 \\ 3\beta_1 + \beta_2 + 3\beta_3 = 14 \end{cases}$$

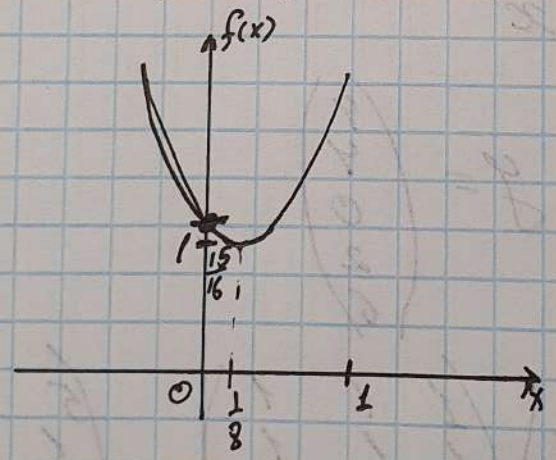
$$\left(\begin{array}{ccc|c} 5 & 1 & 3 & 16 \\ 1 & 3 & 1 & 2 \\ 3 & 1 & 3 & 14 \end{array}\right) \sim \begin{array}{c} \begin{array}{ccc} -7 & -1 & 3 \end{array} \\ \left(\begin{array}{ccc|c} 0 & -14 & -2 & 6 \\ 1 & 3 & 1 & 2 \\ 0 & +81 & 0 & -81 \end{array}\right) \end{array} \sim$$

$$\sim \left(\begin{array}{ccc|c} 0 & 0 & +1 & +4 \\ 1 & 0 & 1 & 5 \\ 0 & 1 & 0 & -1 \end{array}\right) \sim \left(\begin{array}{ccc|c} 0 & 0 & 1 & 4 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \end{array}\right)$$

$\beta_1 = 1 \quad \beta_2 = -1 \quad \beta_3 = 4$

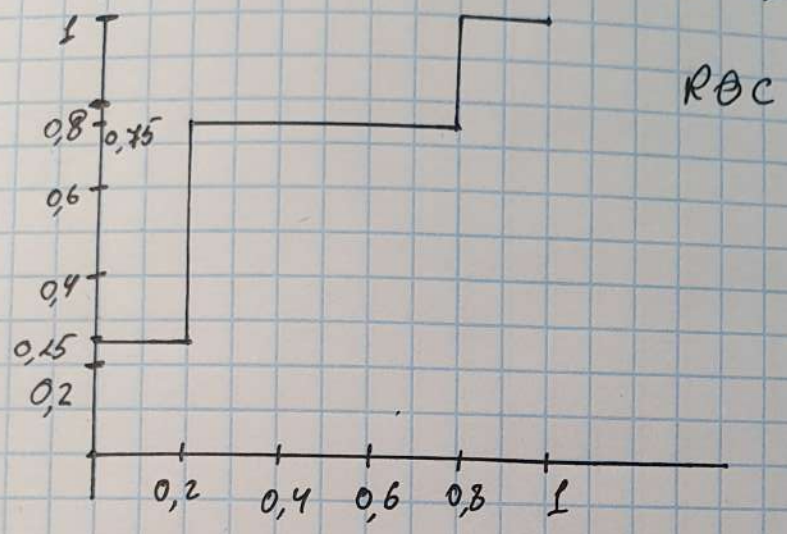
$$\beta = \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix}$$

$f(x) = 1 + (-1)x + 4x^2$ - МНК



№42

i	1	2	3	4	5	6	7	8	9
$y^{(i)}$	0	0	0	0	0	1	1	1	1
$g(x^{(i)})$	0,75	0,15	0,11	0,23	0,09	0,10	0,66	0,82	0,5.



$AUC = 0,7.$

$f(x) = I(g(x) \geq 0,5)$

$f(x^{(i)})$	1	0	0	0	0	0	1	1	1

	$f(x)=0$ pred class=0	$f(x)=1$. pred class=1
$y=0$ True cl=0	$TN=4$	$FP=1$
$y=1$ True cl=1	$FN=1$.	$TP=3$

$FPR = \frac{FP}{TN+FP} = \frac{1}{5}$ $\quad TPR = 1 - FNR = \frac{3}{4}$

$FNR = \frac{FN}{FN+TP} = \frac{1}{4}$ $\quad TNR = 1 - FPR = \frac{4}{5}$

$PPV = \frac{TP}{TP+FP} = \frac{3}{4}$ $\quad error = \frac{FP+FN}{TP+TN+TP+FN} = \frac{2}{9}$

$accur = \frac{TP+TN}{TP+TN+FP+FN} = \frac{7}{9}$ $\quad F_1 = \frac{2}{\frac{1}{PPV}+\frac{1}{TPR}} = \frac{3}{4}.$